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Balancing Weights for Estimating Treatment Effects in Educational Studies

Luke Keele^a , Eli Ben-Michael^b , Matthew Lenard^{c*}, and Lindsay Page^d

^aDepartment of Surgery, University of Pennsylvania, Philadelphia, PA, USA; ^bDepartment of Statistics, Carnegie Mellon University, Pittsburgh, PA, USA; ^cGraduate School of Education, Harvard University, MA, USA; ^dDepartment of Education, Brown University, RI, USA

ABSTRACT

Inverse probability weights often are used in education to estimate causal effects in observational studies. A common practice is to estimate the probability of treatment (e.g., by logistic regression) and use the estimated treatment probabilities to estimate the weights. However, this approach can fail to lead to weights that adequately balance covariates between the treatment and control groups. Balancing weights provide one alternative method for estimating inverse probability weights that does not suffer from this limitation. Balancing weights are based on an algorithm that directly targets reducing imbalances during the estimation process, rather than improving model fit. Here, we provide an introduction to balancing weights methods. We outline the basic mechanics for the estimation of balancing weights, including the specification of a hyperparameter. We review the role of the hyperparameter and detail methods for data driven selection. We compare and contrast the analytic workflow for balancing weights with the more standard inverse probability weighting methods. We present a comparison based on a simulation study to illustrate conditions under which balancing weights perform as well as or better than weights estimated via logistic regression. We conclude with an original empirical application where we evaluate the effectiveness of Pre-K school programs in Wake County, North Carolina.

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1. Introduction

While evidence from randomized trials is considered the gold standard for judging the effectiveness of educational interventions, due to some combination of practical and ethical limitations, researchers often need to conduct observational studies where treatments are non-randomly allocated to the study subjects. It is well understood that when a treatment is non-randomly allocated, differences in outcomes between the treated and control groups may be a function of the pretreatment characteristics of the groups rather than any effect of the treatment itself (Cochran & Chambers 1965; Rubin, 1974). Given this reality, analysts typically apply methods of statistical adjustment to make treated and control groups comparable on observed covariates before

CONTACT Luke Keele  luke.keele@gmail.com  Department of Surgery, University of Pennsylvania, Philadelphia, PA, USA.

*Additional affiliation: Florida State University, Tallahassee, FL, USA.

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estimating causal effects. In observational studies in education, a common strategy is to use inverse propensity score (IP) weighting to render the treated and control groups comparable on observed covariate distributions.

Weighting methods have a long history in survey sampling and causal inference (Horvitz & Thompson, 1952; Lohr, 2009; Robins & Rotnitzky, 1995). The goal is to weight target groups so that they have a similar distribution on a given set of pretreatment covariates. The re-weighted groups can then be directly compared on some outcome of interest, since the re-weighted groups are comparable on baseline characteristics. Weighting estimators are based on the estimated propensity score, and if the estimated propensity score converges to the true propensity score, then the treated and control means will converge to the true means via the balancing property (Robins et al., 2000). Typically, IP weights are based on estimates from a logistic regression model that models treatment take up as a function of baseline characteristics. Weights based on logistic regression, however, may be unstable if that estimate is inaccurate, e.g., due to poor finite-sample overlap between the covariate distributions or small sample size (Ben-Michael & Keele, 2023; Chattopadhyay et al., 2020). Unstable weights result in inconsistent treatment effect estimates (Crump et al., 2009; Kang & Schafer 2007; Li et al., 2018).

Recently, researchers have proposed a class of weighting methods that are designed to target covariate balance directly in the estimation process (Hainmueller, 2012; Zubizarreta, 2015). These methods, known as *balancing weights* methods, provide an alternative estimation procedure for IP weights, fit via a loss function that guarantees covariate balance (Chattopadhyay et al., 2020; Wang & Zubizarreta, 2019; Zhao, 2019; Zhao & Percival, 2017). These methods often outperform IP weights estimated from logistic regression, sometimes have double-robustness guarantees when using the weights alone, and can be de-biased with an outcome model to give double-robustness properties and enable asymptotically valid inference in high dimensional or non-parametric settings (Ben-Michael et al., 2024; Ben-Michael & Keele, 2023; Chattopadhyay et al., 2020; Hirshberg & Wager, 2021; Wang & Zubizarreta, 2019; Zhou et al., 2020).

In this article, we provide an introduction to balancing weights for estimating the causal effects of an education intervention in the context of an observational study. We provide a non-technical review of the general concepts behind balancing weights. Specifically, we review how balancing weights seek to minimize imbalance subject to a constraint or penalty on the complexity of the weights (or minimize complexity subject to a constraint or penalty on the imbalance). The tradeoff between balance and weight complexity is governed by a hyperparameter that is selected by the analyst. We then present a workflow for a weighting analysis that reviews all of the key steps required. After discussing these points, we present a simulation study to demonstrate the conditions where balancing weights will outperform weights estimated via logistic regression. Finally, we conduct an applied analysis to illustrate how analysts can use these methods in their own research.

2. Weighting Estimators of Causal Effects

2.1. Notation and Estimands

First, we outline our notation. We assume we have a sample of units $i = 1, \dots, n$, drawn i.i.d. from some population. We denote a binary treatment using Z_i where $Z_i = 1$

corresponds to the treated condition, and $Z_i = 0$ corresponds to the control condition; n_1 denotes the number of treated units. We let Y_i denote the observed outcome, and we use the potential outcomes framework, where each unit has two potential responses, $(Y_i(1), Y_i(0))$, corresponding to their response under treatment and control. Then, the observed outcome is $Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$. To reduce notational complexity, we will drop the i subscript when taking expectations. We use X_i to denote a vector of baseline covariates. Throughout, we will make use of the *propensity score*—the probability of receiving treatment conditional on the baseline covariates—which we denote as: $e(x) = P(Z = 1 | X = x)$. The re-weighted estimator for the treatment effect is then based on a difference in weighted averages across the treated and control groups:

$$\hat{\Delta}_w = \frac{\sum_{i=1}^n Z_i Y_i w_i}{\sum_{i=1}^n Z_i w_i} - \frac{\sum_{i=1}^n (1 - Z_i) Y_i w_i}{\sum_{i=1}^n (1 - Z_i) w_i},$$

where $w_1, \dots, w_n$ denote the weights that adjust for imbalances in the observed covariates $X_1, \dots, X_n$.

For $\hat{\Delta}_w$ to be a valid estimate requires a set of identification assumptions. First, we assume that there are no hidden forms of treatment. This implies that for unit i , $Z_i = Z$, $Y_i(z) = Y_i$. Second, we assume that a subject's potential outcome is not affected by other subjects' treatment. These two assumptions together are often referred to as the Stable Unit Treatment Value Assumption (Rubin, 1986). Next, we assume that the treatment assignment is ignorable, conditional on the covariates in X :

$$Y_i(1), Y_i(0) \perp\!\!\!\perp Z_i | X_i$$

We also assume there is an overlap in the covariate distributions between the treated and control groups. However, the overlap assumption depends on the choice of the estimand. Next, we review three key estimands that may be of interest in an observational study and outline the associated overlap assumptions.

The estimand is the causal effect of interest and is often defined as a contrast of potential outcomes or counterfactual states. For weighting analyses, we can express three common estimands as variants of a general form. As outlined by Li et al. (2018), we can write the general estimand for a weighting analysis as:

$$\frac{E[h(X)(E[Y(1) - Y(0) | X])]}{E[h(X)]}.$$

where $h(x)$ is a general weighting function. The specific estimand then depends on the form of the weighting function $h(x)$. When $h(x) = 1$, the estimand is the average treatment effect (ATE). When $h(x) = e(x)$, the estimand corresponds to the average treatment effect on the treated (ATT). Finally, when $h(x) = e(x)(1 - e(x))$, the estimand is the average treatment effect for the overlap population (ATO). The ATE measures the average difference in outcomes when all individuals in the study population are assigned to treatment versus when all individuals are assigned to control. The ATT, on

the other hand, measures the average difference in outcomes for those individuals in the population that were actually exposed to the treatment. The ATO captures the treatment effect for the marginal population that might or might not receive the treatment of interest rather than a known, a priori well-defined population such as the treated group. See Greifer and Stuart (2021) for a detailed comparison of these estimands.

Critically, each estimand invokes different overlap assumptions. An overlap assumption ensures that the sample contains information about *both* potential outcomes for all the relevant joint values of the covariates X_i and the treatment Z_i . Under treatment ignorability, each estimand restricts the range of the propensity score $e(x)$ in different ways. The overlap assumption required for the ATE is that no units are always or never treated, i.e., $0 < e(x) < 1$ for each x in the population. For the ATT, the overlap assumption only requires that no units are deterministically treated, i.e., $e(x) < 1$ for each x in the population. Intuitively, we can view the ATT estimand as one that discards members of the control group that are too unlike the treated units. As such, the ATT requires information for fewer joint values of X_i and Z_i , and so has a weaker overlap assumption. For the ATO, we only require that $0 < e(x) < 1$ for *some* x in the population. We can view the ATO estimand as one that discards both treated and control units that do not have overlap in their x values. These estimands answer different scientific questions, so investigators must select which to target based on substantive judgments and the plausibility of the overlap assumptions.

2.2. Model-Based Weights

The weights outlined above can be estimated in a variety of ways. The most common method is to estimate the propensity score using logistic regression by fitting the following logistic regression model via maximum likelihood: $e(x) = 1 / (1 + \exp(-X'\beta))$, modeling treatment exposure as a function of baseline characteristics. The coefficients from this model are used to estimate $\hat{e}(x)$ for each unit in the study. Once the propensity scores are estimated, the form of the weights depends on the target causal estimand. When the ATE is the causal estimand, the weights are the inverse probability of receiving treatment or control: $w_i = 1/\hat{e}(x)$ for the treated units, and $w_i = 1/(1-\hat{e}(x))$ for the control units. When the ATT is the estimand, the weights are $w_i = 1$ for the treated units and the odds of treatment, $w_i = \hat{e}(x)/(1-\hat{e}(x))$, for the control units. For the ATO estimand, the weights are $w_i = 1-\hat{e}(x)$ for the treated units and $w_i = \hat{e}(x)$ for the control units.

Under the modeling approach, the coefficient values in the propensity score model are estimated to optimize the fit of the model measured via the likelihood. A key benefit of the modeling approach is that if the propensity score model is correctly specified, then the estimated weights will be consistent for the true IP weights, and the estimated effect will be consistent for the true effect. However, when the ATE or ATT are the target estimand, the modeling approach suffers from some serious drawbacks in finite samples. Weights based on logistic regression may not balance covariate distributions very well, and model-based weights can be unstable, meaning that a small fraction of the IP weights are very large relative to the other weights (Ben-Michael et al., 2024; Chattopadhyay et al., 2020; Crump et al., 2009; Kang & Schafer 2007; Li et al., 2018). Instability frequently arises from the fact that small errors in estimating

the propensity score can produce much larger errors when estimating the inverse of the weight. This issue is particularly acute when there is poor overlap and there are propensity scores close to either 0 or 1.

Ben-Michael et al. (2021) provide a useful example of how estimation errors in propensity scores can be particularly problematic. Imagine two units in the data that both have true propensity score values of 0.05. For these two units, we make errors in the estimation of 0.04, so that one estimated propensity score is 0.01 and the other is 0.09. The unit with an estimated propensity score of 0.01 will be nine times more influential than the second unit with an estimated propensity score of 0.09. As such, it will be far more influential in the analysis. Instability issues in the estimated propensity scores will be exacerbated when overlap is poor since the relatively few observations in the area of poor overlap can receive very large weights (Crump et al., 2009; Li et al., 2018). Large weights can lead to biased estimates of treatment effects (Kang & Schafer 2007). Instability can also be a result of the fact that logistic regression models are not resistant to outliers (Pregibon, 1982). As such, while weights based on logistic regression may perform well in some applications, using logistic regression may result in weight instability.

2.3. Balancing Weights

Next, we provide a review of balancing weights that is non-technical and focuses on intuition. The primary goal is to find weights that minimize some measure of bias while recognizing that bias reduction comes at a cost in terms of weight complexity. In this literature, the term weight complexity is used to refer to instances where a small fraction of the weights are very large, or, similarly, instances where the distribution of the weights has a large measure of statistical dispersion (Chattopadhyay et al., 2020). Broadly, we can write the balancing weights optimization problem as one where the balancing weights are the solution to minimizing an objective function with three parts:

$$\min[\text{Bias} + \text{Hyperparameter} \times \text{Weight Complexity}]$$

The main component of the objective function is the bias term; we seek to find the set of weights that minimizes this term. While we could *only* minimize bias, the objective function also includes a penalty based on weight complexity. The hyperparameter term serves as a tuning parameter that controls the tradeoff between these two components of the objective function. Estimating propensity score weights via this type of objective function is equivalent to estimating propensity scores via a regularized method of moments estimator, which is one alternative to maximum likelihood estimators (Ben-Michael et al., 2021). This objective function can be implemented in a variety of ways. Various proposals include entropy balancing weights (Hainmueller, 2012), and stable balancing weights (Zubizarreta, 2015), among others (e.g., Ben-Michael et al., 2023; Chan et al., 2016; Tan, 2020).

Next, we write the objective function in a more specific form:

$$\min[\zeta(\text{Imbalance}_M) + \text{Hyperparameter} \times \xi(\text{Weights})].$$

The objective function now include four components: (i) a measure of covariate imbalance (Imbalance_M); (ii) a summary of covariate imbalance ($\zeta(\cdot)$); (iii) a measure of the complexity of the weights ($\xi(\cdot)$); and (iv) the hyperparameter that controls the trade off between bias and weight complexity. These terms determine the parameterization of the weights and the loss function they minimize. Different approaches to balancing weights make different choices about these functions, but they all follow the same principle of minimizing a combination of imbalance and weight complexity. Note that while we have written this objective function as a *penalized* optimization problem—we want to minimize the imbalance while penalizing the weight complexity—we can also equivalently characterize this as a *constrained* optimization problem, as we discuss below. We now review how these terms can be operationalized.

2.3.1. Measuring Covariate Imbalance

First, the bias is operationalized as the imbalance between treated and control covariates, and we index the imbalance by M to denote that there are different modeling choices here. The simplest model is one that is linear and additive; in this case the imbalance measure is simply the difference in (weighted) means between the treated and control group for each covariate. There are two primary ways to complicate this model specification. First, one can add interactions between variables. Second, one can add terms to account for possible nonlinearity in continuous covariates. This can be done for example by including polynomial terms in the model or incorporating spline terms (see Keele, 2008, for an introduction). Including interactions or non-linear terms enriches the model space, and so the resulting imbalance metrics will also include the differences in the (weighted) moments between the treated and control group (e.g., the second moment if using quadratic terms, cross moments if using interaction terms, and complex non-linear moments if using spline terms). This component of the objective function is identical to the functional form selected when logistic regression is used for estimating propensity scores. That is, analysts can complicate the specification of this logistic regression by adding interactions or extra terms to account for possible nonlinearity.

In general, the bigger the model is, the more difficult it is to balance everything in it. For example, if there are 10 covariates, it will tend to be more difficult to balance all 45 possible pairwise interactions between all these variables versus just balancing the 10 covariates alone. The transformations of the covariates that enter into the model for the balance term is often referred to as basis functions. Correct specification of the basis functions is critical to eliminating bias from functional form misspecification (Huang et al., 2022; Setodji et al., 2017). Specification of the basis function can be based either on applying substantive knowledge or using machine learning methods (Ben-Michael et al., 2024). To use an ML-based approach, one first randomly samples a small percentage of the data. With this sample, one can use a generalized additive model to identify which variables have nonlinear associations with the outcome and random forest models to identify key interactions. This information is used to specify the basis function in the remaining data, while the data used for these models is discarded from the analysis. Recent work has also explored how to use kernel methods to fully automate model specification (Hazlett, 2020; Huling & Mak,

2020; Wong & Chan, 2018) as well as using sieve-type methods (Wang & Zubizarreta, 2019) and boosting approaches (Zhao, 2019).

2.3.2. Summarizing Covariate Imbalance

Once one has decided what features of the covariate distribution to balance, the next question is how to summarize the overall level of imbalance across those features via the $\zeta(\cdot)$ function, which represents some form of aggregation. The choice of aggregation measure affects the resulting optimization problem. For example, Zubizarreta (2015); Wang and Zubizarreta (2019), Athey et al. (2018), and Tan (2020) choose $\zeta(\cdot)$ to measure the maximum imbalance across all of the features of the covariates using the L^∞ norm.¹ This controls the imbalance in the least-balanced covariate after weighting, and aligns with the common heuristic that covariates not be imbalanced by more than 0.1 standard deviations. Ben-Michael et al. (2023) instead use the sum of the squared imbalances across the features of the covariates by using the L^2 norm. Like the L^∞ norm, targeting this measure of imbalance prioritizes the covariates that are highly imbalanced, but is smoothed in such a way that a small amount of imbalance in many covariates can outweigh a large amount of imbalance in one covariate. The choice of imbalance function is closely related to regularization approaches for generalized linear models. On the one hand, the imbalance function is linked to constraints placed on the complexity of the outcome model (Hirshberg et al., 2019; Hirshberg & Wager, 2021). For example, if one assumes that the outcome model is sparse, the corresponding aggregation function is the L^∞ norm; if instead the outcome model coefficients has a bounded L^2 norm, then the corresponding aggregation function is the L^2 norm. See Ben-Michael et al. (2021) for further discussion.

2.3.3. Measuring Weight Complexity

Next, the dispersion function $\xi(\cdot)$ is used to operationalize a measure of weight complexity. Typically, weight complexity is measured via the dispersion of the weights. A variety of ways have been used to measure this dispersion. For instance, several approaches measure the dispersion via entropy $w(\log w - 1)$ (e.g., Hainmueller, 2012). Optimizing for the entropy measure can be shown to be equivalent to estimating a logistic regression model for the propensity score with a calibrated loss function (Tan, 2020; Zhao, 2019; Zhao & Percival, 2017). Other approaches use the sum of the squared weights (e.g., Ben-Michael et al., 2023; Hirshberg et al., 2019; Zubizarreta, 2015). The sum of the squared weights is related to the inverse of the *effective sample size* of the weighting estimator, and so minimizing the sum of the squared weights maximizes the effective sample size. Finally, the dispersion function can include additional constraints on the weights, in particular constraining the weights to be non-negative. This constraint prevents extrapolation beyond the support of the data and is included in a

¹Zubizarreta (2015) and Wang and Zubizarreta (2019) first scale the features so that imbalance in more important features can be reduced more than imbalance in less important features. We can also characterize this as changing the choice of the aggregation function to be a weighted norm, e.g. with a matrix of weights V serving as additional hyperparameters.

variety of approaches (e.g., Athey et al., 2018; Ben-Michael et al., 2023; Hainmueller, 2012; Wang & Zubizarreta, 2019; Zubizarreta, 2015).

2.3.4. Choosing the Hyperparameter

Last we turn to the hyperparameter term, which controls the tradeoff between bias (as measured by the imbalance) and variance (as measured by the weight complexity/dispersion). To start this discussion, we should note that there are alternate, equivalent representations of the optimization problem, each with different hyperparameters. For these various hyperparameter representations we will assume that generally smaller hyperparameter values emphasize reducing imbalance and larger values will place more emphasis on reducing weight complexity. Some implementations reverse this relationship, and this is purely an implementation choice.

One common way to implement the hyperparameter is to write the objective function such that it minimizes the complexity of the weights, subject to a constraint on the balance:

$$\begin{aligned} & \min \xi(\text{Weights}) \\ & \text{subject to } \zeta(\text{Imbalance}_M) \leq \text{Hyperparameter} \end{aligned}$$

In this constrained formulation of the objective function, decreasing the hyperparameter leads to a more stringent requirement on the covariate balance. The extreme case where the hyperparameter is set to be zero leads to an *exact balance* constraint that $\text{Imbalance}_M = 0$.² Balancing weight methods that enforce an exact balance constraint (e.g., entropy weighting (Hainmueller, 2012)), fully prioritize the bias. However, if there are even a moderate number of covariates relative to the sample size, it will typically be impossible to find weights that can exactly balance the covariates, and so the optimization problem will be infeasible (see, e.g., the sample size requirements in Zhao & Percival, 2017). This is more likely to be the case when the specification includes more complex basis functions with interactions and higher order terms (Ben-Michael et al., 2024). Instead, the constraint can be written to enforce *approximate* balance. For example, Zubizarreta (2015), Wang and Zubizarreta (2019) and Tan (2020) (through the Lagrangian dual form) set $\zeta(\cdot)$ to be the (weighted) L^∞ norm and use a non-zero hyperparameter value. Now the optimization problem is designed to find weights of minimal complexity subject to the constraint that the most-imbalanced covariate is at most the value of the hyperparameter.

If we fix the choice of imbalance measure Imbalance_M , aggregation function $\zeta(\cdot)$, and weight complexity function $\xi(\cdot)$, for any choice of hyperparameter in the constrained formulation there exists a choice of hyperparameter in the penalized formulation such that the solutions to the constrained and penalized formulations are the same, as long as the initial hyperparameter is not zero and the constraint is feasible.³

²To avoid additional cumbersome notation, here we are implicitly assuming that the aggregation measure is positive definite, so that $\zeta(\cdot)$ implies that $\cdot$ is typically a norm, which satisfies this property.

³The exact balance case where the hyperparameter is set to zero is an edge case that can be handled differently depending on the implementation. In particular, we can use an alternative implementation of our main specification to

The other common implementation uses a formulation that penalizes the variance of the weights, typically with a single hyperparameter (e.g., Athey et al., 2018; Ben-Michael et al., 2023; Hirshberg & Wager, 2021).

Each implementation has different costs and benefits. For example, the constrained formulation gives a direct interpretation of the hyperparameter in terms of the imbalance constraint, while in the penalized formulation the relationship between the hyperparameter and level of imbalance is opaque. As such, achieving a particular level of covariate balance requires searching over different hyperparameter choices. On the other hand, if the hyperparameter is chosen to be too small in the constrained formulation, then the optimization problem might be infeasible and the underlying optimization would give an error. In comparison, the penalized formulation will always have a unique solution as long as the hyperparameter is non-zero and the complexity measure is strongly convex. Ultimately, the choice between the constrained and penalized formulation depends on which is easier to implement for the particular choice of imbalance measure, aggregation function, and weight complexity function. For example, if we use the variance of the weights as the complexity function and the L^∞ norm as the imbalance summary measure, then with modern quadratic programming solvers it is more straightforward to implement the constrained form of the optimization problem. Computational solutions are more straightforward with the penalized form when using the L^2 norm.

How should one select this hyperparameter? One informal method for selecting the hyperparameter is to calculate the balance and the effective sample size for a range of hyperparameter values. One then selects a value that produces an acceptable level of balance and effective sample size. Below, we review how to calculate the quantities for this type of analysis. However, there are proposals in the literature for data-driven hyperparameter selection. For example, Wang and Zubizarreta (2019) outline a method based on the bootstrap for that chooses a value of the hyperparameter that achieves the best balance on bootstrapped subsamples of the data. While they develop this for the constrained setting it can be adapted to the penalized setting as well.⁴

For the penalized setting with the variance of the weights as the complexity term, Hirshberg and Wager (2021) discuss that the MSE-optimal choice of the hyperparameter is the residual variance of the outcome Y , conditional on the covariates X . Ben-Michael et al. (2024) propose to follow this suggestion by extracting the hyperparameter from a regression of Y on X restricted to the control group ($Z=0$). The hyperparameter is then set to the variance of the residuals from this regression, making the objective function a proxy for the mean-squared error of the weighting estimator. Note that this proposal is specific to a penalized implementation with the variance of the weights as the complexity function, and it is not clear how this approach would generalize to other implementations. Comparisons of these two methods and alternative data-driven methods for hyperparameter selection is an important avenue for future research. We demonstrate both methods in the application below.

instead solve an optimization problem that minimizes the weight complexity and penalizes the imbalance term by a factor of λ , with the convention that $\lambda = 0$ and that $\lambda \rightarrow \infty$ so that the solution to $\lambda \rightarrow \infty$ is to set $\lambda = 0$.

⁴While Wang and Zubizarreta (2019) present an approach with a separate hyperparameter for each covariate, they alter the method to use a single hyperparameter across all covariates when proposing the selection procedure.

2.3.5. Additional Approaches

There are other approaches that are alternatives to IP weights based on logistic regression that do not fall into the structure we define above. For example, there is the method of covariate balancing propensity score weighting (CBPS), which is a combination of logistic regression and balancing weights because it incorporates balance constraints into the estimation of the logistic regression model (Fan et al., 2023; Fong et al., 2018; Imai & Ratkovic, 2014). An alternative approach is overlap weighting (Li et al., 2018); this focuses on the ATO and uses logistic regression to estimate the propensity scores and the corresponding overlap weights. As Li et al. (2018) note, using logistic regression in this way does guarantee balance for the estimated overlap-weighted sample, unlike when estimating the ATE or ATT. Ben-Michael et al. (2024) outline a balancing weights method that targets the ATO.

2.3.6. Estimating Effects after Estimating the Weights

The balancing weight procedures described above only changes the estimation procedure for the IP weights. Once the weights are estimated, treatment effect estimation proceeds in the same way as with model-based weights. That is, the treatment effect estimate can be calculated either as a contrast of weighted means, a weighted regression model, or using a double robust approach. For example, treatment effects can be computed by regressing the outcome on the treatment group indicator using weighted least squares, weighting by the estimated weights. Such regression models are often referred to as marginal structural models (Hernán & Robins, 2020; Robins et al., 2000). The marginal structural model framework can be adapted to balancing weights by replacing the model-based weights with the estimated balancing weights in the weighted least squares model.

2.3.7. Balancing Weights and Double Robustness

One common approach to estimating treatment effects combines a propensity score model with an outcome model, and is referred to as “doubly robust,” since treatment effects are consistently estimated when either model is correctly specified (Scharfstein et al., 1999). In fact, such doubly robust (DR) estimators were first developed as extensions of the IPW estimator (Scharfstein et al., 1999). Next, we discuss how balancing weights can be used to form DR estimators and the specific conditions that need to hold for double robustness.

For some types of balancing weights, the covariate balance constraints provide a type of double robustness guarantee depending on the true outcome and propensity scores models. Critically, this form of double robustness holds without the need for an outcome model. For example, when estimating the ATT with entropy balancing weights with an exact balance constraint (Hainmueller, 2012), the estimate will be consistent if either (a) the true propensity score is a logistic regression model with the basis functions that are used to measure imbalance_M, or (b) the true outcome model is linear in those basis functions (Källberg & Waernbaum, 2023; Zhao & Percival, 2017). This is because entropy balancing is a consistent estimator for the IP weights, regardless of the outcome model, and exact balance ensures that the weighted contrast will be an unbiased estimate for the ATT under the linear model, regardless of the propensity score model. This double robustness property can also hold even if we can only achieve

approximate balance because either the covariates are high dimensional or if we use a large set of transformations of the covariates. For example, Wang and Zubizarreta (2019) and Hirshberg et al. (2019) consider using sieve-type and kernel methods, respectively. They show that in certain asymptotic regimes it is possible to achieve good enough balance in a broad enough set of transformations of the covariates so that double-robustness holds even for a nonparametric outcome model and propensity scores.

However, it is not always the case that approximate balancing weights without an outcome model will have a double robustness property, since approximate balance could lead to some amount of bias. A more general approach to double robustness is to “augment” the balancing weights (e.g., Athey et al., 2018; Hirshberg & Wager, 2021) by incorporating a model for the outcome along the same lines as in the “augmented” inverse propensity score weighted (AIPW) estimator for model-based IP weights (Robins et al., 1994; Scharfstein et al., 1999). Like AIPW estimators, augmented balancing weights estimators have double-robustness properties under various regularity conditions, including that the IP weights and the outcome model can be consistently estimated. Furthermore, like AIPW, augmented balancing weights can often satisfy *rate double-robustness* properties in high dimensional or non-parametric settings, so that the estimates are asymptotically normal under certain conditions⁵ In fact, augmented balancing weigh approaches can even allow for consistent and asymptotically normal estimation when the propensity score cannot be consistently estimated, because they can also be viewed as de-biasing outcome model-only estimates of the treatment effect (Athey et al., 2018; Hirshberg & Wager, 2021).

In the analysis that follows, we use the balancing weights from Ben-Michael et al. (2023). As we outlined above, this implementation of balancing weights uses the L^2 norm of the vector of covariate imbalances, and uses the average of the squared weights as the dispersion measure, constraining the weights to be non-negative. Here, we briefly consider the double robustness properties of this method of balancing weights. We show in the [supplementary materials](#) that this implementation of balancing weights is a special case of the kernelized approach in Hirshberg et al. (2019), using a kernel defined by the basis functions. As such, if the basis functions correctly specify either the outcome model or IP weights, then these balancing weights may have the double robustness property without further augmentation via an outcome model, depending on the assumptions about the asymptotic regime. Moreover, if we augment the weights with an outcome model, then this type of balancing weights is a special case of the augmented minimax linear estimator proposed by Hirshberg and Wager (2021). Therefore, it is likely that the augmented weights will have a double robustness property under weaker conditions, though this still requires the basis functions to be correctly specified. These result suggest that the balancing weights in Ben-Michael et al. (2023) enjoy the same double robustness properties as those in Hirshberg et al. (2019) and Hirshberg and Wager (2021). However, formal verification of these properties is not yet available in the literature and is beyond the scope of this article.

⁵Specifically, the estimates are asymptotically normal if the product of the rate of decay of the estimation error for the outcome model and the level of covariate balance is faster than root- n .

3. Workflow

Next, we review what we call the workflow for an analysis using balancing weights. By “workflow,” we refer to all the analytic steps that should occur in an observational study that employs balancing weights. Most of the steps in the workflow are identical to those for an analysis based on model-based IP weights. Specifically, we recommend the following steps: (1) select covariates, (2) select the basis function, (3) estimate the weights, (4) perform diagnostic checks, (5) estimate treatment effects, and (6) perform sensitivity analysis. Next, we review each of these steps while noting how these steps differ depending on whether one is using model-based or balancing weights.

3.1. Estimating the Weights

The first two steps in the workflow are identical regardless of which types of weights are used. First, the analyst selects the set of covariates that are thought to render the treated and control groups exchangeable. Second, one needs to specify the basis function in terms of selecting interaction terms or nonlinear terms. If one were using model-based weights, the basis function would comprise the specification of the logistic regression model used to estimate the weights. For balancing weights, the basis function is used in the objective function to define the imbalance measure. After doing this, we estimate the weights, which includes selecting the hyperparameter.

3.2. Diagnostic Measures

Once the weights are estimated, little differs in the remaining steps of the workflow. We recommend three diagnostics: (1) inspecting the distribution of the weights for large values, (2) conducting balance checks, and (3) computing effective sample size. For inspection of the weights, the goal is to identify large weights. There are various rules of thumb for how to define large weights (Cole & Hernán, 2008). For example, weights larger than 10 are often flagged as large. Next, we should find that the weighted treated and control differences in the baseline covariates should be small, i.e., the weights should balance the treated and control distributions. The standardized difference is a common statistic used to measure imbalance before and after matching or weighting. For covariate k , the standardized difference before weighting is

$$\hat{\Delta}_k = \frac{\bar{X}_{tk} - \bar{X}_{ck}}{\sqrt{(V(X_{tk}) + V(X_{tc}))/2}}$$

where X_{tk} and X_{ck} are the treated and control group vectors for covariate k , and $V(X_{tk})$ and $V(X_{tc})$ are the variances for the treated and control group. The standardized difference after weighting is

$$\hat{\Delta}_{wk} = \frac{\bar{X}_{wtk} - \bar{X}_{wck}}{\sqrt{(V(X_{tk}) + V(X_{tc}))/2}}$$

where $\bar{X}_{wtk}$ and $\bar{X}_{wck}$ are weighted means based on the estimated balancing weights. Traditional balance diagnostics require covariate-by-covariate comparisons. It is also useful to calculate an overall balance metric. One such metric is the percentage of bias reduction (PBR). To define this, we use $\hat{\Delta}_{uw}$ to refer to the vector of standardized differences for all K covariates in the unweighted data, and $\hat{\Delta}_w$ for the vector of standardized differences for all K covariates in the weighted data. Using these two quantities, we the PBR is:

$$PBR = 100\% \times \left[\frac{1}{K} \sum_k \left| \hat{\Delta}_w \right| / \frac{1}{K} \sum_k \left| \hat{\Delta}_{uw} \right| \right].$$

In sum, the PBR is a measure of how much a set of weights improves balance across all the covariates compared to the baseline balance. With model-based IP weights, if there are large imbalances after weighting, options are limited. With balancing weights, one changes the value of the hyperparameter for additional bias reduction. For both model-based IP weights and balancing weights, if there are large imbalances even after making adjustments to the weights, one could change the estimand to the ATO. Note that Ben-Michael and Keele (2023) show that when overlap is poor, balancing weights often still perform well, allowing analysts to focus on the ATT rather than having to use the ATO estimand.

Another important diagnostic is the effective sample size. The effective sample size captures the fact that some units receive zero or nearly zero weight and thus contribute little to the analysis. It also captures the loss of information due to the fact that some units receive weights larger than one. The effective sample size (ESS) after weighting is

$$n^{\text{eff}} \equiv \left(\sum_{i=1}^n \hat{w}_i \right)^2 / \sum_{i=1}^n \hat{w}_i^2.$$

If the outcomes are homoskedastic with variance σ^2 , the standard error of the weighting estimator will be $\sigma / \sqrt{n^{\text{eff}}}$, and so the effective sample size serves as a proxy for the standard error. Note that if the ATT is the estimand, we only calculate the ESS for the control group.

Beyond just being diagnostic measures, PBR and ESS can be used for hyperparameter selection. One can calculate the ESS and PBR for a series of hyperparameter values and plot them against each other. One then finds an inflection point where the drop in PBR is relatively large and the gain in ESS is small. Iterating through multiple values of the hyperparameter is permissible, since it does not rely on any outcome information.

3.3. Estimating the Effects

Once the diagnostics are complete, the final step is to estimate the treatment effect. As we noted above, one can use differences in weighted means, a marginal structural model, or an augmented balancing weights estimator. Variance estimation for the treatment effect is somewhat more subtle. In the marginal structural model framework, when the ATE is the target estimand, for model-based IP weights, the asymptotic

variance for IPW estimators is estimated by using the Huber-White sandwich estimator of the variance in the weighted outcome model (Pirracchio et al., 2016; Ramsey et al., 2019). When the ATT is the target estimand, alternative methods based on stacked estimating equations are necessary (Reifeis & Hudgens, 2022). For balancing weights, the Huber-White sandwich estimator can be used for both the ATE and the ATT but is conservative (Ben-Michael et al., 2024). Ben-Michael et al. (2024) outline a less conservative approach, but find that, in terms of coverage, both methods produce similar results. Both methods are available in the `balancer` package in R that we use in the analyses presented below (Ben-Michael et al., 2024). See the [supplementary materials](#) for additional details on this form of balancing weights.

3.4. Sensitivity Analysis

The final step of the analytic workflow is to conduct a sensitivity analysis. We provide a brief review of sensitivity analyses, but readers should see Rosenbaum (2010) for more details and Page et al. (2020) for a detailed example in the education research context. In a sensitivity analysis, we seek to *quantify* the degree to which a key assumption must be violated for an original conclusion to be reversed. In the context of an observational study, we probe the assumption that there are no unobserved confounders. Soriano et al. (2023) develop sensitivity analysis methods for balancing weights. This form of sensitivity analysis is consistent with that in Rosenbaum (2010) which relies on the Γ parameter to ask what level of confounding would reverse study conclusions. In this form of sensitivity analysis, Γ represents the effect of an unobserved confounder on the odds-ratio scale. If we set $\Gamma=1$, that is consistent with the assumption that the unobserved confounder has no effect, while $\Gamma>1$ represent larger amounts of unobserved confounding. We can summarize the sensitivity analysis with the Γ changepoint—the Γ value at which the bounds on the treatment effect estimate include zero. If this Γ value is large, we can conclude that inferences are insensitive to hidden bias related to unobserved characteristics. If the Γ value is small, it suggests that inferences are vulnerable to hidden confounders. Next, we conduct a simulation study to further explore how model-based IP weights compare to balancing weights.

4. Simulation

We conduct two simulation studies to investigate how well balancing weights methods perform relative to model-based IP weighting estimated via logistic regression in finite samples. For both methods, we target the ATT estimand. We estimate balancing weights using the `balancer` library in R (Ben-Michael et al., 2024) which implements the methods outlined in Ben-Michael et al. (2023).

4.1. Simulation 1

For the first simulation, we first describe the data generating process (DGP). For this DGP, there are five observed covariates that are independent draws from a multivariate standard Gaussian distribution: $X = (X_1, \dots, X_5)^\top$. The treatment indicator

Z_i is generated from the following model $Z_i = \mathbb{1}(Z^* > 0)$ where $Z^* = (1.5X_1 + 1.5X_2 + .7X_1X_2)/c + \text{Unif}(-0.5, 0.5)$. In this DGP, we use c to control the overlap between the treated and control groups. Smaller values of c will reduce the level of overlap between the treated and control groups. Next, we generate potential outcomes under $Z_i = 0$ as: $Y(0) = X_2 + X_3 + \epsilon_1$ and under $Z_i = 1$ as: $Y(1) = Y(0) + \delta$, where ϵ_1 is drawn from a standard Normal distribution and δ represents the treatment effect. In the simulations, we set $\delta = 5$. In sum, the outcome model is linear, while the propensity score model contains an interaction.

In this simulation, we make the following set of comparisons. First, we estimate treatment effects using IP weights based on logistic regression. Next, we estimate treatment effects using balancing weights. Next, we use IP weights based on a random forest to estimate the propensity scores (Welz et al., 2024). Note that we use the default hyperparameter setting of 500 trees. We refer to these as RF IP weights. For all three methods, we use two different specifications. The first specification is linear and includes no higher order terms, and the second specification includes both linear and quadratic terms for all five covariates. We include these two specification to understand the consequence of over-specification in a setting where the basis is linear. That is, we should expect some increase in variance, but the extent of the increase in variance is not well understood. Note that none of the methods include any adjustment via an outcome model.

In these simulations, we vary the overlap in the covariate distributions. Specifically, we use values of 1, 2.5, 5, 7.5, and 10. When $c=1$, overlap is poor, and when $c=10$, overlap is good. The performance of the weighting estimators should degrade as overlap gets weaker. Results in Ben-Michael and Keele (2023) suggest balancing weights should outperform model based weights when overlap degrades. It is not clear *a priori* how the random forest-based IP weights will perform as overlap changes.

In this set of simulations, we set the value for the hyperparameter using the data-driven method outlined above where we use the variance of the residuals from a regression of the outcome on the covariates in the control group. See the [supplementary materials](#) for an additional simulation study on the performance of this method of hyperparameter selection. For the simulations, we set the sample size to 1000 and use 1000 Monte Carlo replications. We measured absolute bias using the absolute value of the standard metric of the average difference between the true treatment effect and the estimated treatment effect. We record absolute bias, since the direction of the bias differs by estimation method. We also measure the root mean-squared error (RMSE) using the standard formula of the root of the average squared difference between the true treatment effect and the estimated treatment effect.

Figure 1 contains the results for this simulation. Across all levels of overlap, we find that model-based IP weights generally outperform the RF IP weights, and balancing weights have the best performance compared to the other two alternatives. In the low overlap condition, model-based IP weights with the simple specification has the worst performance, which is consistent with other research in the extant literature. However, once overlap decreases further, the bias for model based IP weights increases dramatically as does the RMSE. However, the model-based weights never outperform balancing weights. The more complex specification does not change the performance

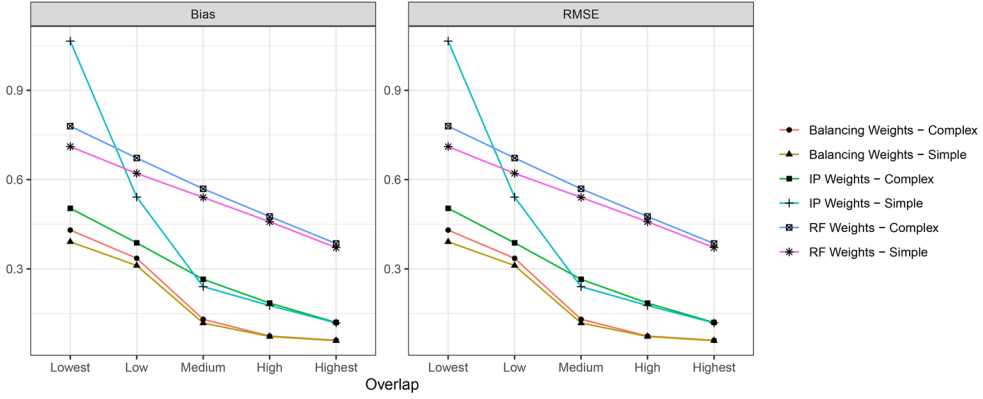


Figure 1. Absolute bias and Mean Square Error of model-based IP weights, balancing weights, and RF IP weights, varying the degree of covariate overlap.

of either balancing weights or the RF IP weights. In this case where the outcome model is a simple linear function, using a standard, flexible ML method does not improve the performance of the model-based weights. Perhaps better tuning of the hyperparameter for the random forest might improve performance. However, little guidance exists on how to set the RF tuning parameter in this setting.

We performed one additional simulation using this framework, where we omit a key variable. We would expect omitting a key variable to increase the bias for all the methods. The results from the simulation are consistent with our expectations. Full results from this simulation can be found in the [supplementary materials](#).

4.2. Simulation 2

In the second simulation, we consider a more complex specification. Again, there are five observed covariates $X_1, \dots, X_5$. Covariates X_1, X_2, X_3 are drawn from a mean zero multivariate Normal distribution with the following variance-covariance matrix:

$$\begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -0.5 \\ -1 & -0.5 & 1 \end{bmatrix}$$

Covariates X_4 and X_5 are mutually independent of each other and X_1, X_2, X_3 . In the data-generating process, $X_4 \sim \text{Unif}[-3, 3]$ and $X_5 \sim \text{Bernoulli}(0.5)$.

The treatment indicator Z_i is generated from the following model $Z_i = \mathbb{1}(Z^* > 0)$ where $Z^* = (1.5X_1 + 1.5X_2 - 2X_3 + X_4 + 0.5X_5) / c + \text{Unif}(-0.5, 0.5)$. Next, we generate potential outcomes under $Z_i = 0$ as: $Y(0) = X_1^2 + X_2^2 + X_3^2 + \epsilon_1$ and under $Z_i = 1$ as: $Y(1) = Y(0) + \delta$, where ϵ_1 is drawn from a standard Normal distribution and δ represents the treatment effect. In the simulations, we set $\delta = 5$. As before, we use c to control the overlap between the treated and control groups, and smaller values of c will reduce the level of overlap between the treated and control groups. Again, we use values of 1, 2.5, 5, 7.5, and 10. In this set of simulations, we set the value for the hyperparameter using the data-driven method outlined above.

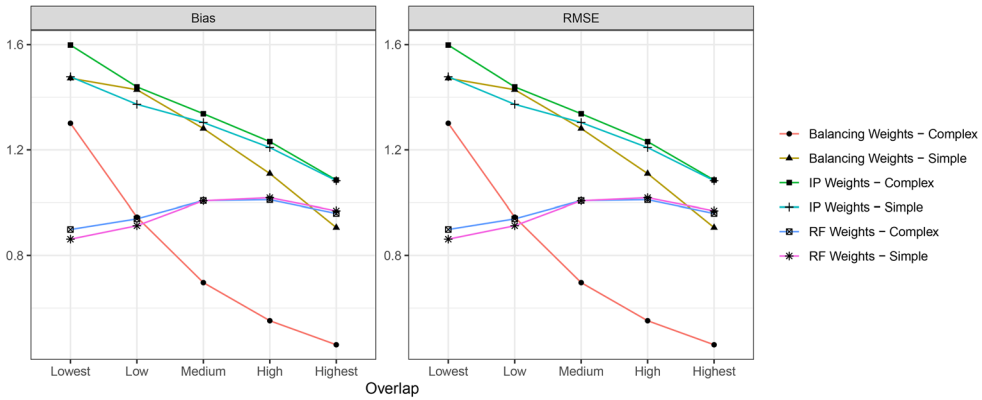


Figure 2. Absolute bias and Mean Square Error of model-based IP weights and balancing weights, varying the degree of covariate overlap.

In this simulation, we make the same set of comparisons as in the first simulation study: model-based weights, balancing weights and RF IP weights. Again, we use each method with either a linear or linear and quadratic specification. RF weights are of interest in this setting, since the RF model may be able learn the basis function even when the quadratic specification is not included. Again, we measure absolute bias and RMSE as performance metrics, and we set the sample size to 1000 and use 1000 Monte Carlo replications.

Figure 2 contains the results for this second simulation. When overlap is poor, balancing weights and model-based weights methods are biased by a similar magnitude, however, balancing weights with the more complex specification have the smallest bias among these two methods. As overlap improves, the performance of both balancing weights and model-based weights improves. In this setting, the difference between model-based IP weights with and without the complex specification is smaller. However, the complex specification for balancing weights results in much better performance. We also find that for the two conditions with the highest overlap, balancing weights with the simple specification outperforms logistic-based IPW methods with the more complex specification. As expected, the performance of RF weights do not differ by specification. As such, RF IP weights have the smallest bias and RMSE when overlap is lowest. However, the performance of the RF weights does not improve with overlap which means that balancing weights outperform the RF weights in three of the five scenarios.

4.3. Summary

The results from our simulation reveal a few key points about the relative performance of model-based IP weights versus balancing weights. In the first simulation, when the outcome model has a simpler functional form, we see that the level of overlap has a key effect on the performance of the two methods. In settings where the overlap is very good, the performance of the two methods will be nearly the same; there are many contexts where the two methods will produce the same results. However, as we observed, balancing weights are far more robust in settings where the overlap degrades.

In scenarios where the overlap is poor, balancing weights still perform well, while model-based IP weights suffer from higher bias and much higher mean-squared errors.

Next, we considered the case where the outcome model has a more complex functional form. In this case, there are two key results. First, the results emphasize that the more complex specification is critical to improve the performance of the balancing weights. That is, with the complex specification balancing weights outperformed all the other methods, even when the outcome is misspecified. However, when overlap is quite poor, none of the methods can overcome the misspecification of the outcome model. When overlap is at least moderate, balancing weights with the simpler specification outperformed model-weight IP methods. Of course, adding an outcome model would also be an avenue for further bias reduction.

5. Empirical Application: Wake County Pre-K Programs

Social scientists have long argued for investments in the development of young children through pre-kindergarten (pre-K) programs prior to the start of formal schooling (e.g., Duncan & Magnuson, 2013; Heckman & Masterov, 2007). In response, an extensive body of evidence has been compiled on the effectiveness of pre-K programming (Cascio & Schanzenbach, 2013; Duncan & Magnuson, 2013; Gray-Lobe et al., 2022; Li et al., 2020; Lipsey et al., 2018). To demonstrate the use of balancing weights in an empirical application, we evaluate voluntary pre-K in the Wake County Public School System (“Wake County”). Wake County offers pre-K services through three different programs. Students can enroll in NC Pre-K, a statewide program started by the state in 2001 (Carr et al., 2021; Little et al., 2019). Students can also apply for pre-K services through Title I or Head Start. Each of the three programs has different eligibility requirements: while NC Pre-K and Head Start are income based, Title I pre-K is based on students’ academic and developmental needs. Our study represents the most recent and largest test of pre-K impacts in Wake County.

In Wake County, students are given the Kindergarten Initial Assessment (KIA) within the first week of kindergarten enrollment by the classroom teacher in collaboration with the student and a parent/guardian. The KIA includes a question asking whether the incoming kindergarten student was enrolled in a pre-K program within the past year. Since the district does not maintain administrative data on student enrollment in specific pre-K programs, we use the responses to this question for the treatment variable of interest. Note that given this measurement strategy, our sample excludes any students that used one of the three public pre-K options but then enrolled in private educational institutions. We merged the KIA data with Wake County administrative records, which include time-invariant student characteristics such as sex, race/ethnicity, school assignment, and special status characteristics. These data are collected within 20 days of the start of kindergarten, and we assume that gender and racial markers are stable from the time when students were enrolled in pre-K within a year prior. Since the district’s administrative records include few pretreatment control variables, we also include U.S. Census tract and zip-code level indicators made publicly

available through Opportunity Insights (OI) (Chetty et al., 2018, 2022).⁶ The OI data provide pretreatment (2010 Census) measures of economic and social capital and mobility for each student in our data. Finally, we collect student math and reading performance at the end of grade 3, which is the first year of accountability testing. Since our entering kindergarten class enrolled in fall 2015, we use summative test scores for the last pre-pandemic academic year, 2018–2019, which is when our kindergarten cohort would be enrolled in grade 3.

5.1. Data Analysis

In the data, 7898 students attended a pre-K program and 2596 did not. Given this distribution, we define treatment as having not attended pre-K. We also conduct a subgroup analysis for children with limited English proficiency. We do this to investigate whether the effect of pre-K is different within this subgroup. Notably, overlap may be more limited between the treated and control groups in such subgroups. In our analysis, we target the ATT estimand. In the [supplementary materials](#), we include a software tutorial with example code for a publicly available data set. In that analysis, we target both the ATT and ATO.

Here, we follow the workflow outlined above. We start by specifying the basis function to include nonlinear terms and interactions between variables. First, we took a random sample of 10% from the control group. We use this subset of the data as a training sample. Using this training sample, we first seek to identify nonlinearities. To that end, we fit a generalized additive model between the outcome and all the continuous control variables. Based on the results from this model, we identified three variables to include as nonlinear: incarceration percentage, U.S. Census form return percentage, and volunteering percentage. Next, we used the training sample to identify interactions. Random forest models have metrics that allow the analyst to compute variable importance metrics for interactions (Inglis et al., 2022). We fit a random forest using the training sample, and we then calculated the variable importance metrics for the interactions. Based on these results, we identified four interactions between Black and median income, Black and college education percentage, Black and U.S. Census form return percentage, and white and median income. As such we will seek to balance not just the full set of baseline covariates, but also the higher order terms for the three nonlinear variables and the four interaction terms. Note that we discard the training sample, and do not use it for any additional analyses.

The next step in our analysis is selection of the hyperparameter for the balancing weights. In our analysis, we used the bootstrap method from Wang and Zubizarreta (2019) as well as the method based on the outcome. Using the bootstrap method, we estimated a hyperparameter value of 0.01. Using the outcome based method, we estimated a hyperparameter value of 0.73 for one of the outcomes and 0.74 for the other outcome. Given the nearly identical values, we use the larger of these values to estimate one set of weights for both outcomes. Next, we estimated the weights using the balancer package. We estimated two sets of weights for both the bootstrap and outcome

⁶Opportunity Insights data and corresponding documentation are available at <https://opportunityinsights.org/data>.

based hyperparameters. We then calculated the PBR for both sets of weights. The PBR values for both hyperparameters were identical to the second decimal place. As such, the different hyperparameter values do not translate into meaningful differences in terms of the ability to balance baseline covariates. Hereafter, we report results for the weights with the larger hyperparameter.

Next, we estimated a set of weights using logistic regression. After estimating the weights, we compare the effective sample sizes. After discarding the training sample, we have 7,108 controls and 2596 treated units. Since we are targeting the ATT estimand, the treated sample size is fixed. For the model-based IP weights, the ESS for the controls is 2208 and for balancing weights it is 2628. As such, we retain a bit more sample size under balancing weights. It is also useful to inspect the distribution of the weights to ensure that none of the weight values are extreme. The median weight is 0.38 for the balancing weights with a maximum weight of 2.7. The median weight is 0.32 for the model-based IP weights with a maximum weight of 4.9. Here, the model-based weights do not produce any extreme weights.

Next, we evaluate the performance of the weights in terms of how well they balance the distribution of baseline covariates. [Figure 3](#) provides a comparison of the balance before and after weighting via balancing weights. The mean differences before weighting between the pre-K and non-pre-K cohorts are quite large: a number of these standardized differences are greater than 0.50. However, after weighting the two groups are nearly identical in terms of the key covariates. In fact, the PBR for the balancing weights is 99%.

Next, we compare the model-based IP weights to the balancing weights. [Figure 4](#) plots the balance for both sets of weights. In this plot, it is clear that the model-based IP weights also produce small imbalances. The PBR for the model-based weights is 95%. However, the balancing weights are able to nearly exactly balance the covariate distributions. In sum, consistent with the simulations, in a context with good overlap, balancing weights improves on the performance of model-based IP weights but the difference between the two methods is small.

Next, we estimate the treatment effect of not having attended pre-K. Following our outline above, we simply regress the outcomes on the treatment indicator for having not attended pre-K using a weighted regression model. We used the same method of analysis for both the balancing weights and the model-based IP weights. We first report results for the balancing weights. We find that students that did not attend pre-K had significantly lower test scores in both math and reading. For reading, test scores were lower by 0.16 standard deviations (95% CI: $-0.21, -0.11$). For math, test scores were lower by 0.15 standard deviations (95% CI: $-0.20, -0.10$). Results based on model-based IP weights were quite similar with test scores lower by 0.13 standard deviations for both reading and math. Notably, the larger effective sample size for balancing weights results in confidence intervals that were shorter by approximately 12%. We also estimate the treatment effects using an augmented balancing weights estimator. These estimates are almost identical to those reported above (-0.157 and -0.151 , respectively).

We conduct separate sensitivity analyses for math and reading scores. For reading test scores, the changepoint value for Γ is 1.21, which results in bounds on the estimate of -0.20 and 0.005 . For math scores, the changepoint value for Γ is 1.22 which results in

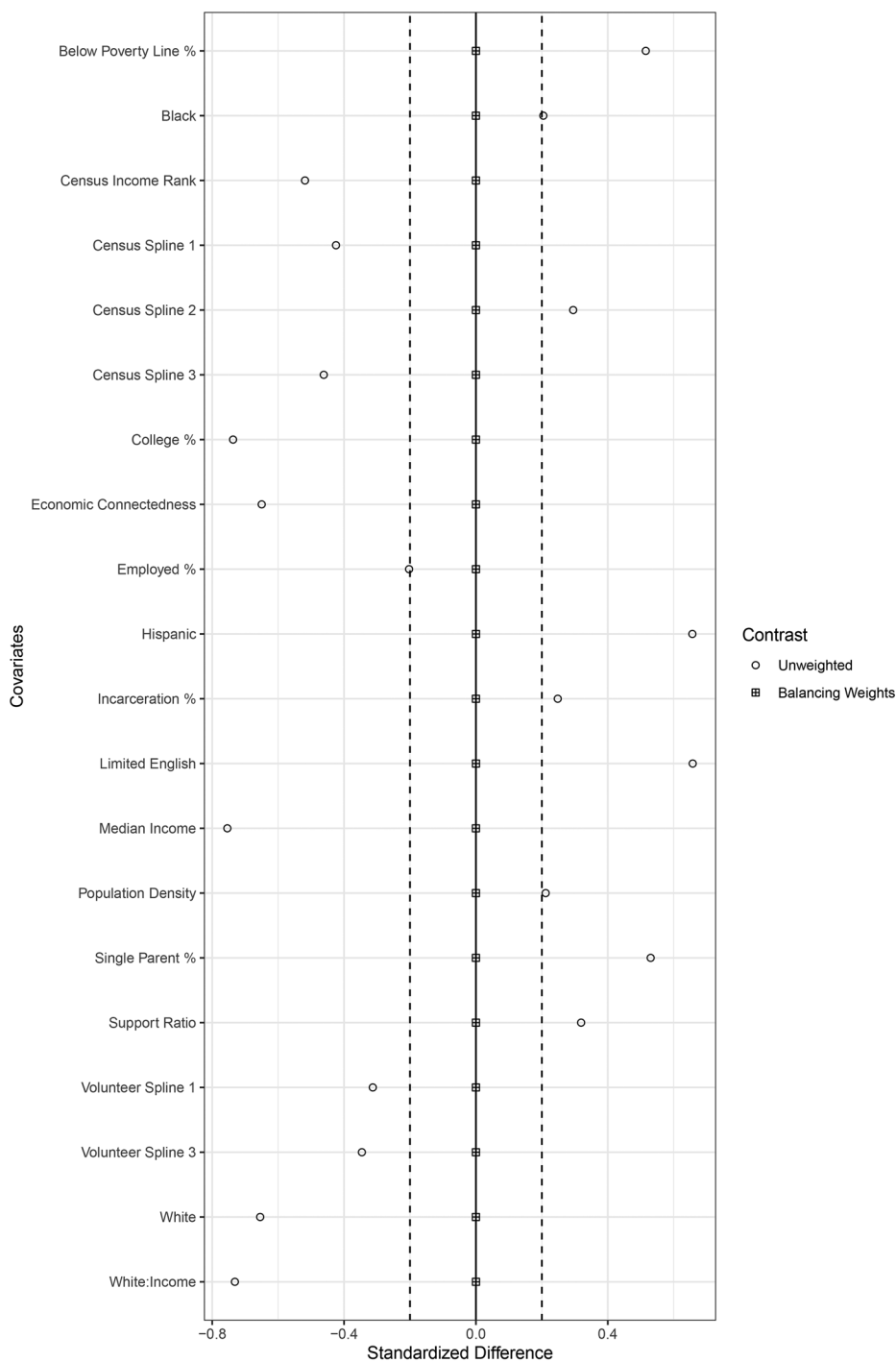


Figure 3. Balance before and after weighting for the covariates with the largest baseline imbalances using balancing weights. Dashed lines represent 0.20 and -0.20 standard differences.

bounds on the estimate of -0.29 and 0.007 . Not surprisingly, given the similarity of the point estimates, the sensitivity analysis results are nearly identical for both subjects. Since Γ is on an odds-ratio scale, this implies that if there were a binary unobserved confounder

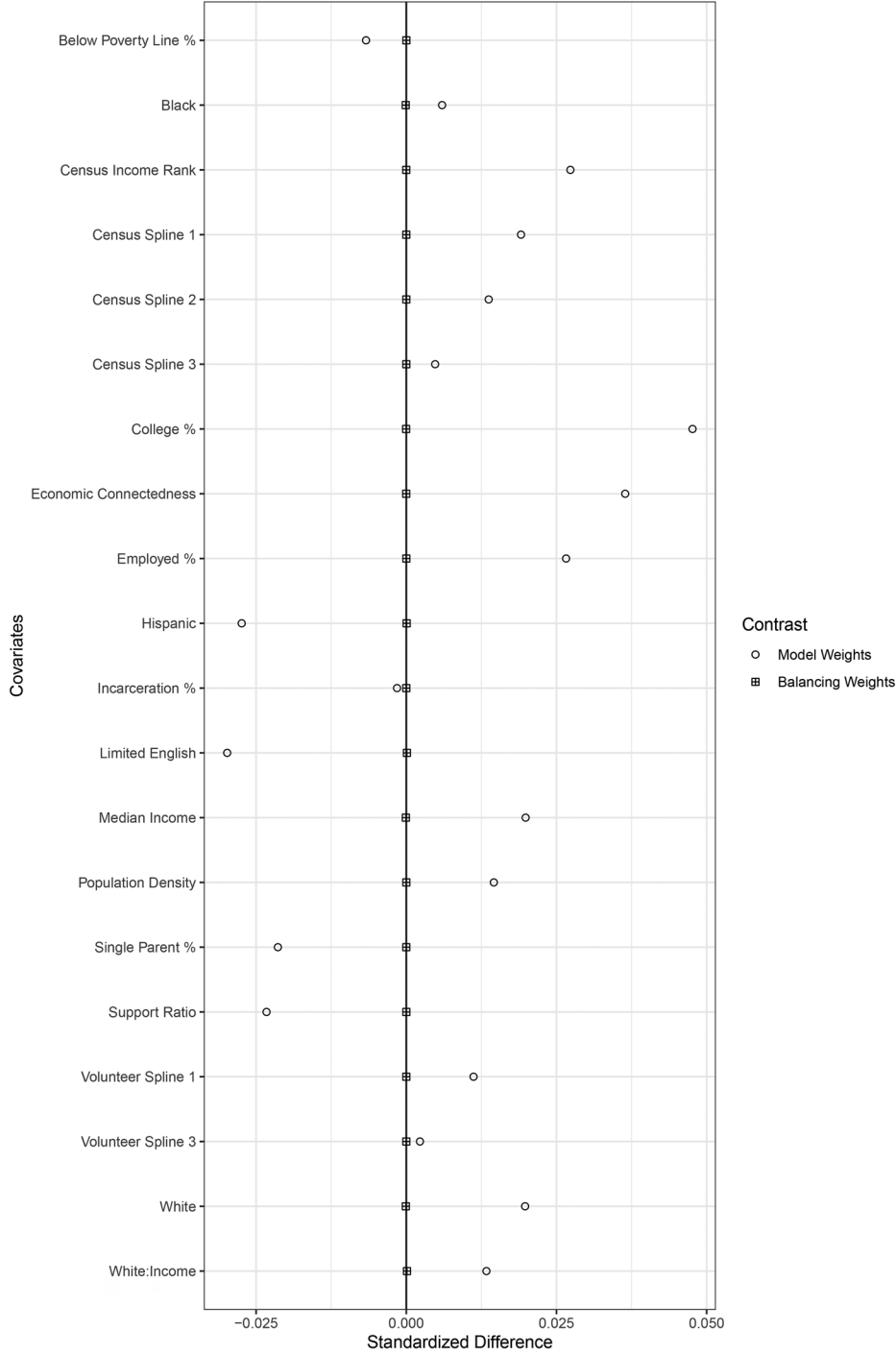


Figure 4. Balance after weighting for the covariates with the largest baseline imbalances comparing model-based IP weights to balancing weights.

that caused the odds of treatment to differ by 21%, this could explain the results in our study. This is a relatively small value of Γ , which implies that the results in our study could be explained by unobserved confounding.

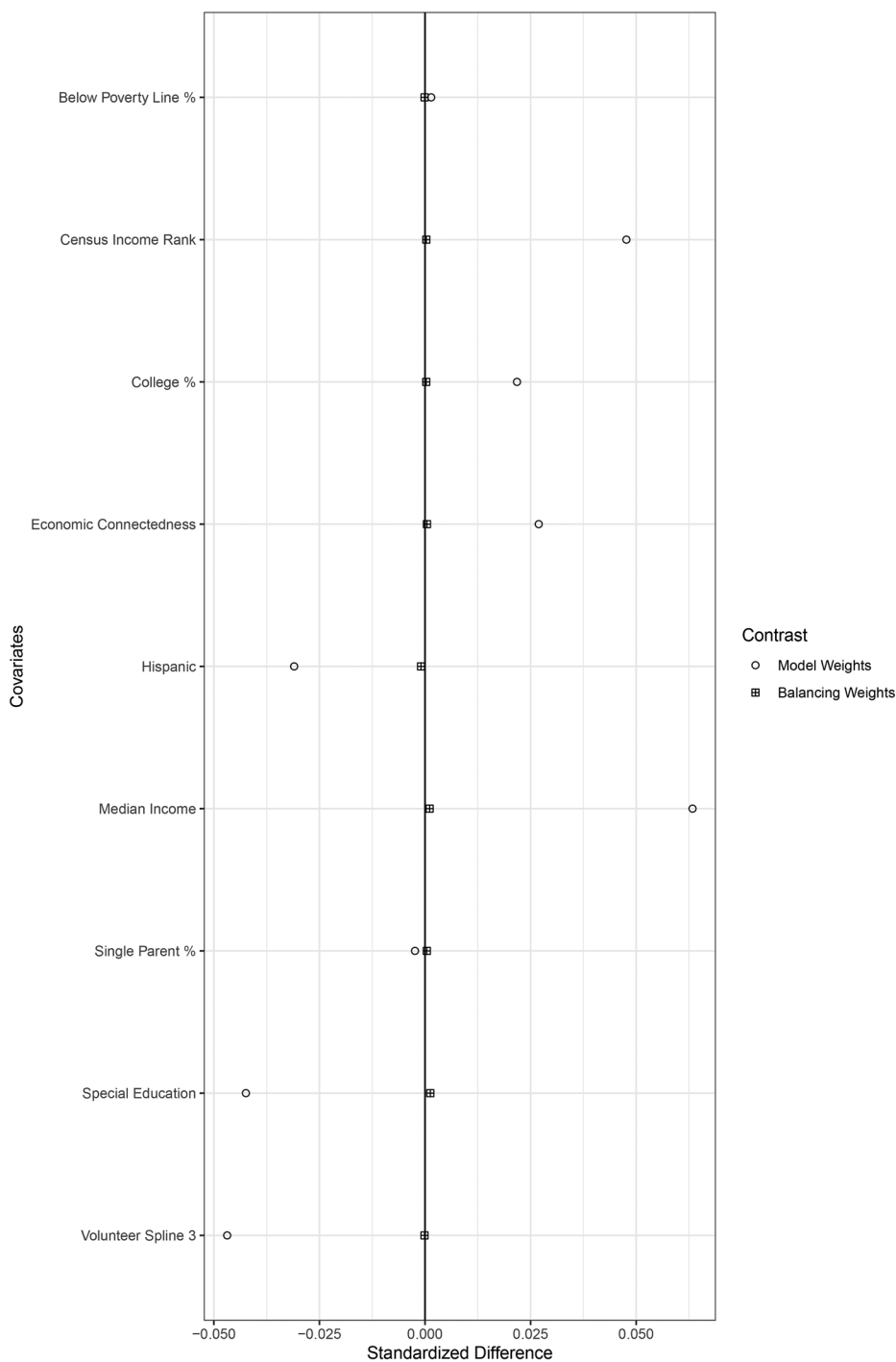


Figure 5. Balance after weighting for the covariates with the largest baseline imbalances comparing model-based IP weights to balancing weights, restricted to children with limited English proficiency.

Next, we present the results for the subgroup analysis. Here, we re-estimate the weights in the subset of students with limited English proficiency. The median weight is 0.83 for the balancing weights with a maximum weight of 5.6. The median weight

is 0.63 for the model-based IP weights with a maximum weight of 17.9. Now, the model-based IP weights are notably larger than the balancing weights. This difference is reflected in how well each method balances the data. For balancing weights, the PBR remains 99%, but for model-based IP weights, the PBR drops to 82%. The differences are readily apparent in [Figure 5](#). Again, the results are consistent with our simulation study. As overlap decreases, the performance of the model-based IP weights is degraded, but the performance of balancing weights is unchanged.

Now we turn to the estimated effect of not having attended pre-K among students with limited English proficiency. Using balancing weights, we find that reading and math test scores were each lower by 0.11 standard deviations. In both cases, the confidence intervals narrowly include zero ($p = .07$). Using model-based IP weights, we find that reading scores were lower by 0.11 standard deviations and math scores were lower by 0.06 standard deviations. As such, the lower levels of imbalance produced by model-based IP weights may have resulted in some bias in the treatment effect estimate for math scores. In addition, for math scores, the confidence interval for balancing weights is nearly 32% shorter in length. Again, estimates with augmented balancing weights were nearly identical.

6. Discussion

Observational studies, while subject to the problem of confounding, remain an important source of evidence on the effectiveness of educational interventions. Currently, model-based propensity score weighting is widely used to adjust for observed confounders in observational studies. As we outlined, in many applications, using logistic regression to estimate propensity score weights can lead to biased estimates, especially if overlap between treated and control units starts to degrade. Recent research has developed an alternative method of estimation known as balancing weights. In this manuscript, we provided an introduction to balancing weights in an educational research context. The key advantage of balancing weights is that they are designed to directly target imbalance with a penalty on the variance of the weights. An expanding literature has demonstrated that balancing weights often outperform IP weights estimated from logistic regression (Ben-Michael et al., [2024](#); [2024](#); Ben-Michael & Keele, [2023](#); Chattopadhyay et al., [2020](#); Wang & Zubizarreta, [2019](#); Zhou et al., [2020](#)). In this article, we add to this literature. In our simulation studies, we showed that when overlap is good, model-based IP weights and balancing weights performed equally well. However, when overlap begins to degrade, balancing weights produced lower bias and mean-squared error. We also found that when the outcome model is nonlinear, balancing weights generally performed better than model-based weights even when the balancing weights are misspecified.

We also reviewed the overall workflow for a weighting analysis. Our workflow outlines what we view as best practice for any weighting analysis. In particular, we provided guidance on specifying the basis function to ensure that it includes key interactions and nonlinear terms. Finally, we then followed this workflow in a study on the effectiveness of pre-K programs in Wake County. In the main analysis, model-based IP weights and balancing weights performed in similar fashion. However,

in a key subgroup, where overlap was not as good, balancing weights outperformed model-based IP weights in terms of balance as measured by PBR. For analysts interested in using these tools, we provide a software tutorial with example code and a publicly available data set in the [supplementary materials](#). The replication materials for the tutorial can be found on our GitHub page at <https://github.com/ljk20/balwts>.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Open Research Statements

Study and Analysis Plan Registration

There is no registration associated with this study.

Data, Code, and Materials Transparency

The data and code associated with this study can be found on GitHub: <https://github.com/ljk20/balwts>. The materials associated with this study are not publicly available.

Design and Analysis Reporting Guidelines

Not applicable.

Transparency Declaration

The corresponding author (the manuscript's guarantor) affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

Replication Statement

This manuscript reports an original study.

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ORCID

Luke Keele  <http://orcid.org/0000-0002-3859-2713>

Eli Ben-Michael  <http://orcid.org/0000-0002-1175-4129>

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